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(54) **METHOD FOR IMPROVING CONTROL GATE UNIFORMITY DURING MANUFACTURE OF PROCESSORS WITH EMBEDDED FLASH MEMORY**

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CPC **H10B 41/30** (2023.02); **H10B 41/40** (2023.02)

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See application file for complete search history.

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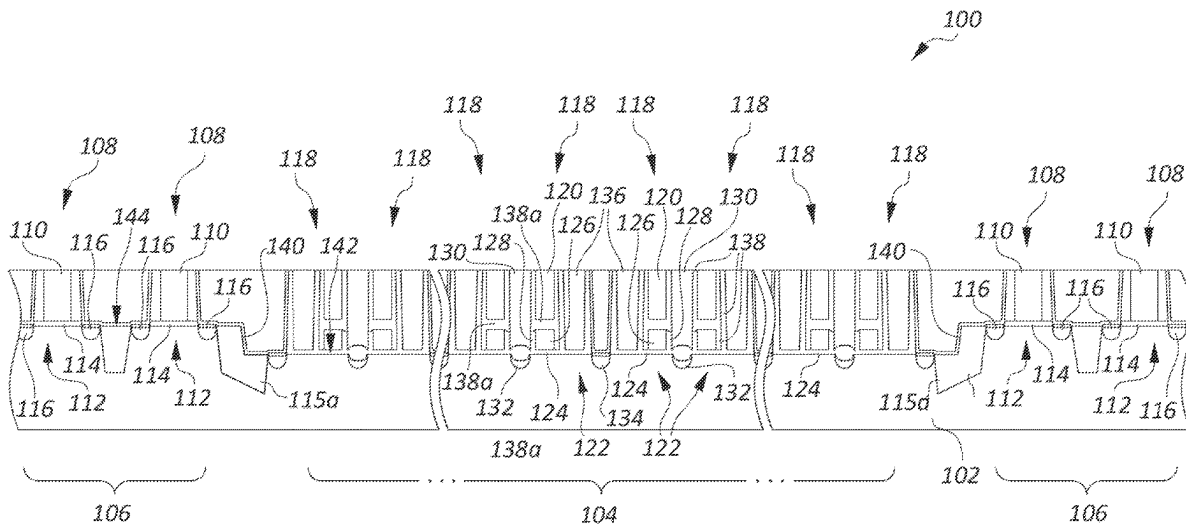
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(57) **ABSTRACT**

A method includes planarizing a protective layer over gate materials overlying a recessed region in a substrate. The planarizing includes forming a first planarized surface by planarizing a sacrificial layer over the protective layer, and forming a second planarized surface of the protective layer by etching the first planarized surface of the sacrificial layer at an even rate across the recessed region. An etch mask layer is formed over the second planarized surface, and control gate stacks are formed in the recessed region by etching the gate materials.

20 Claims, 7 Drawing Sheets



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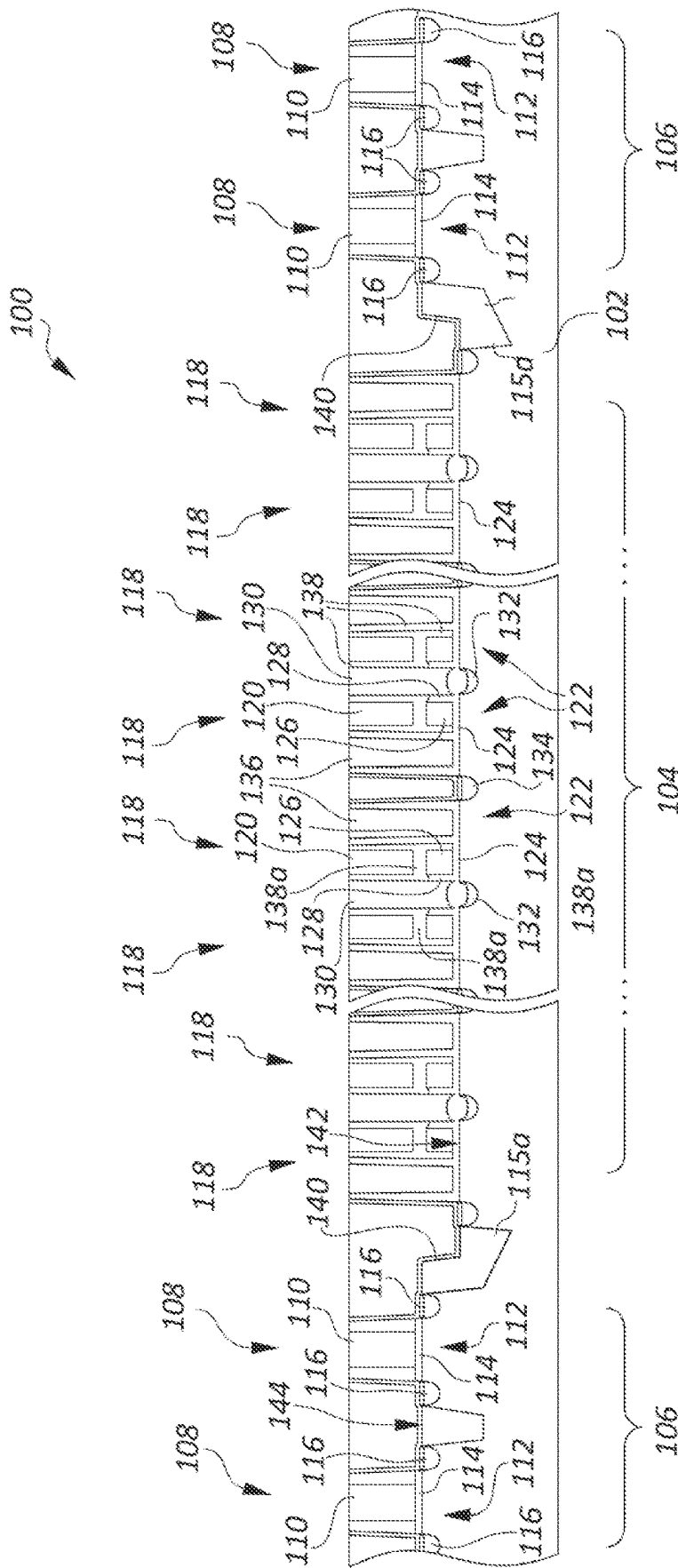


FIG. 1

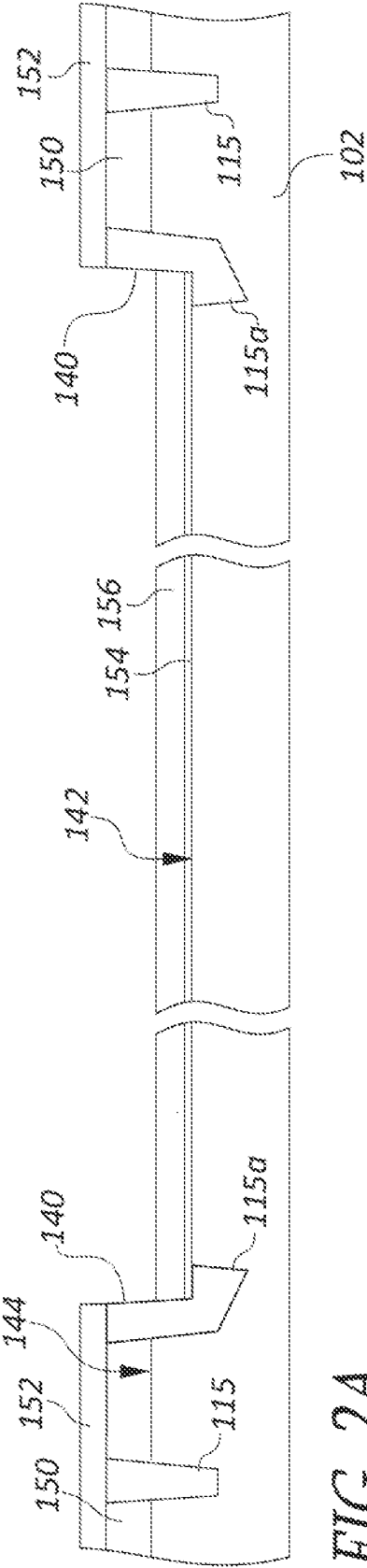


FIG. 2A

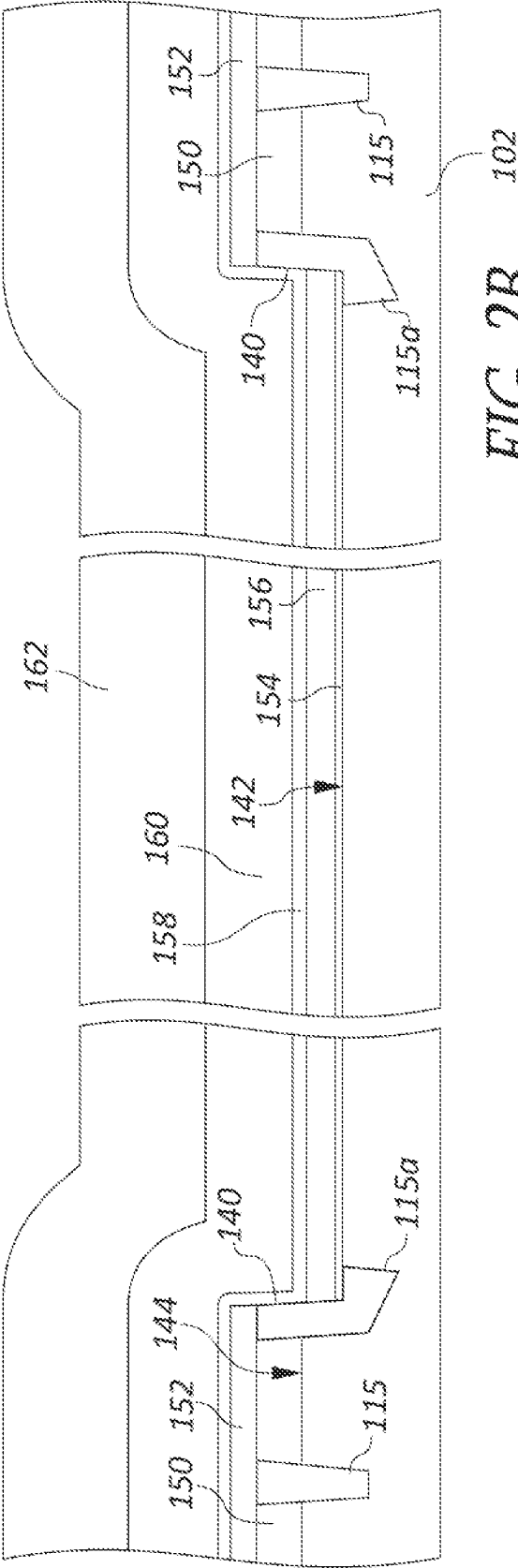
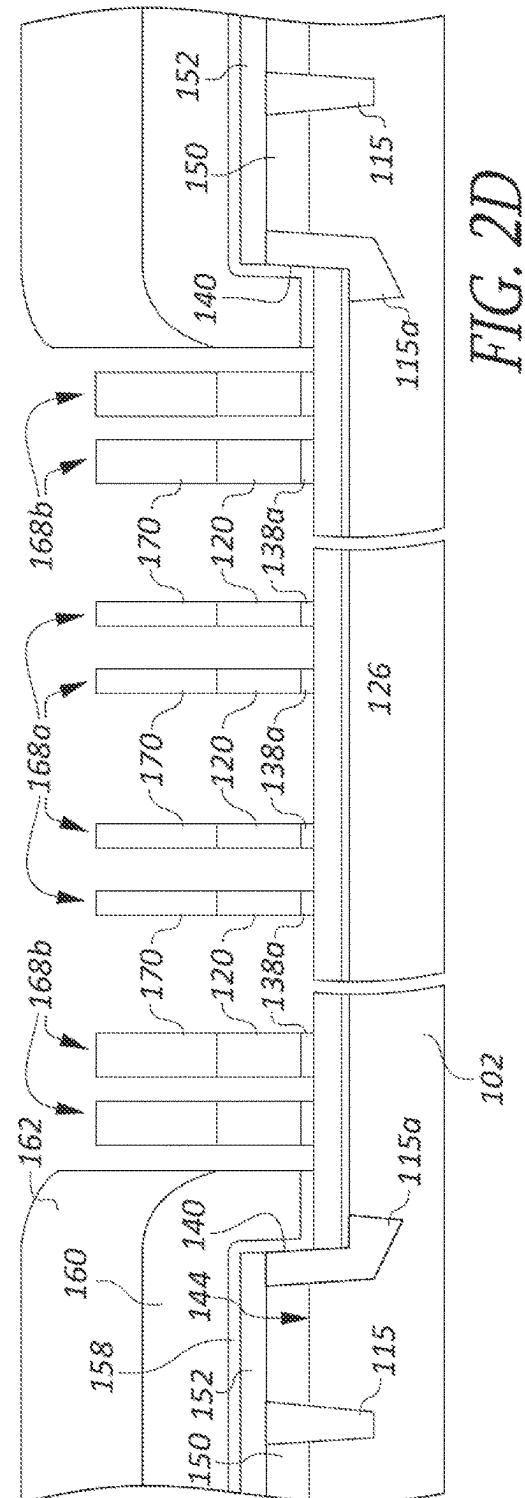
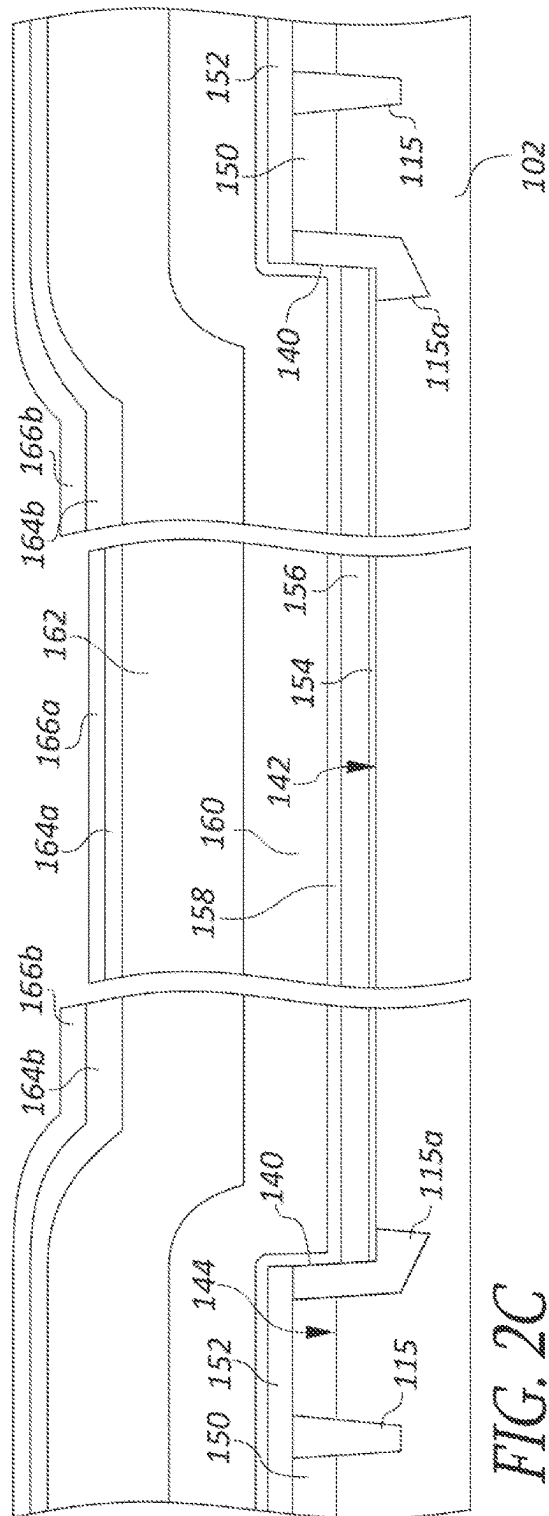
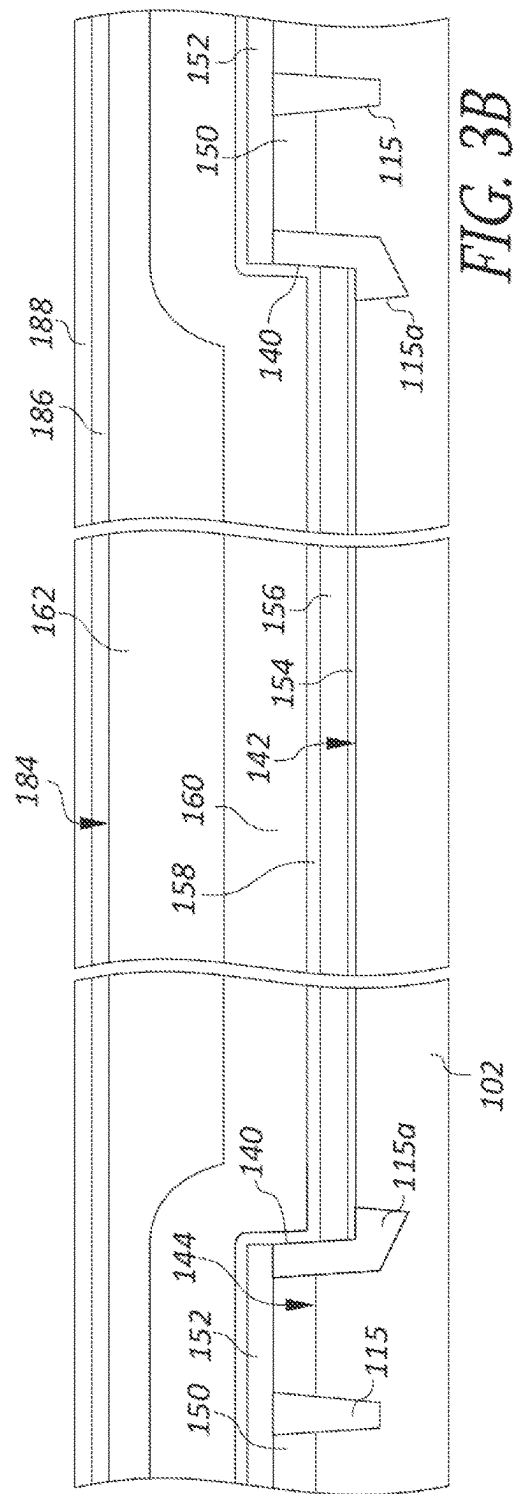
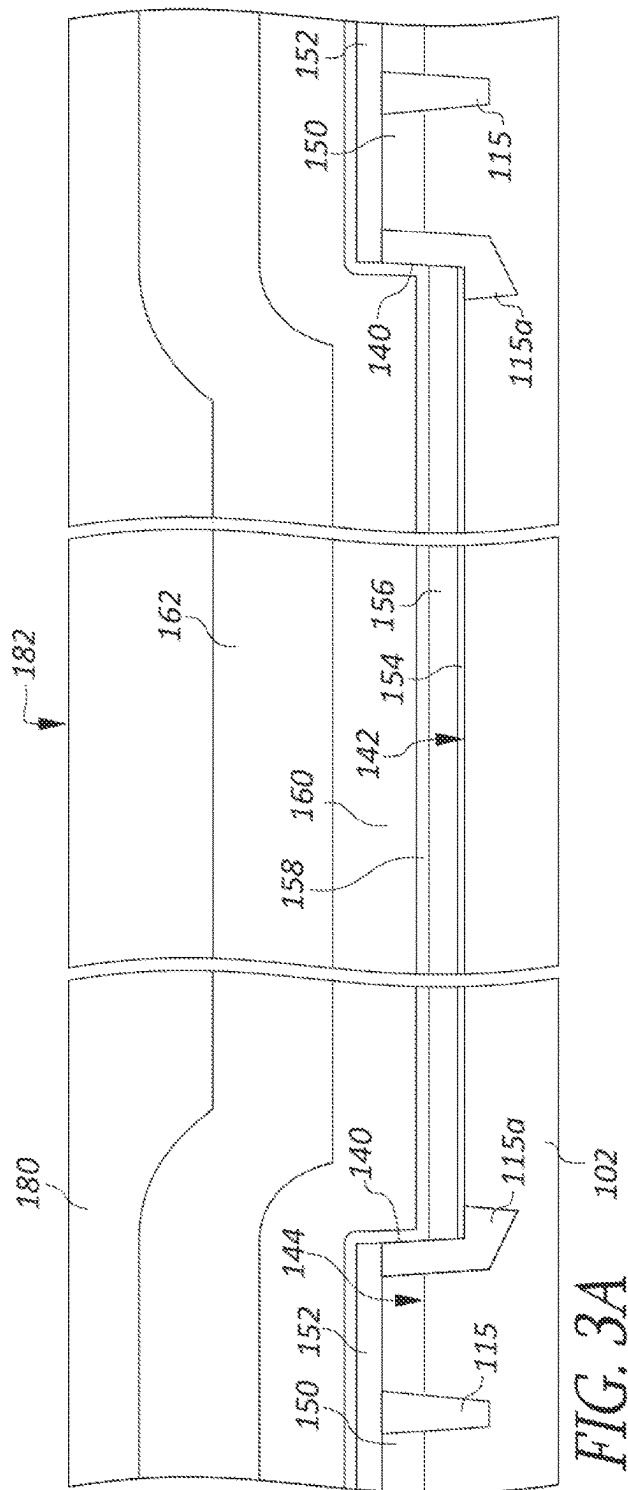


FIG. 2B





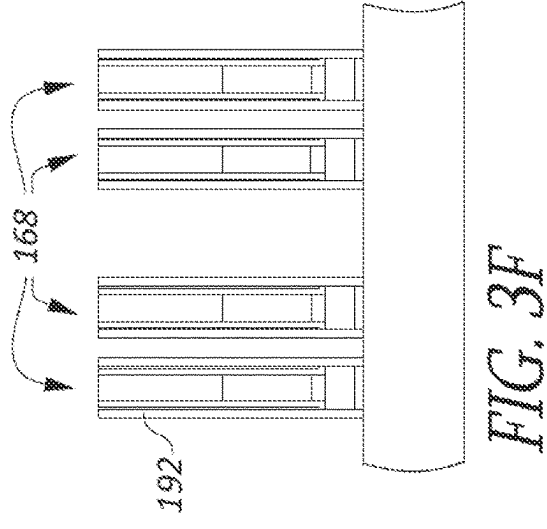
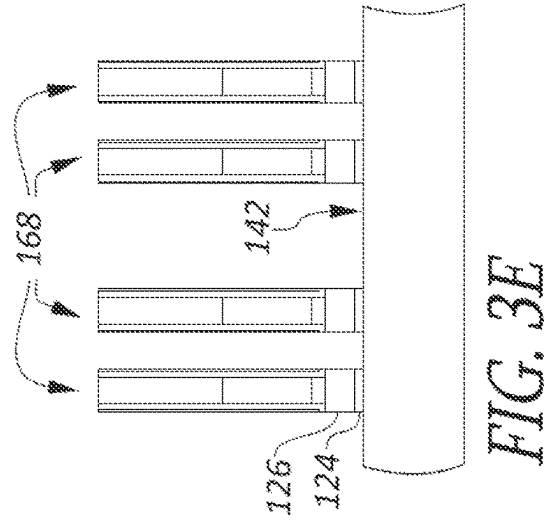
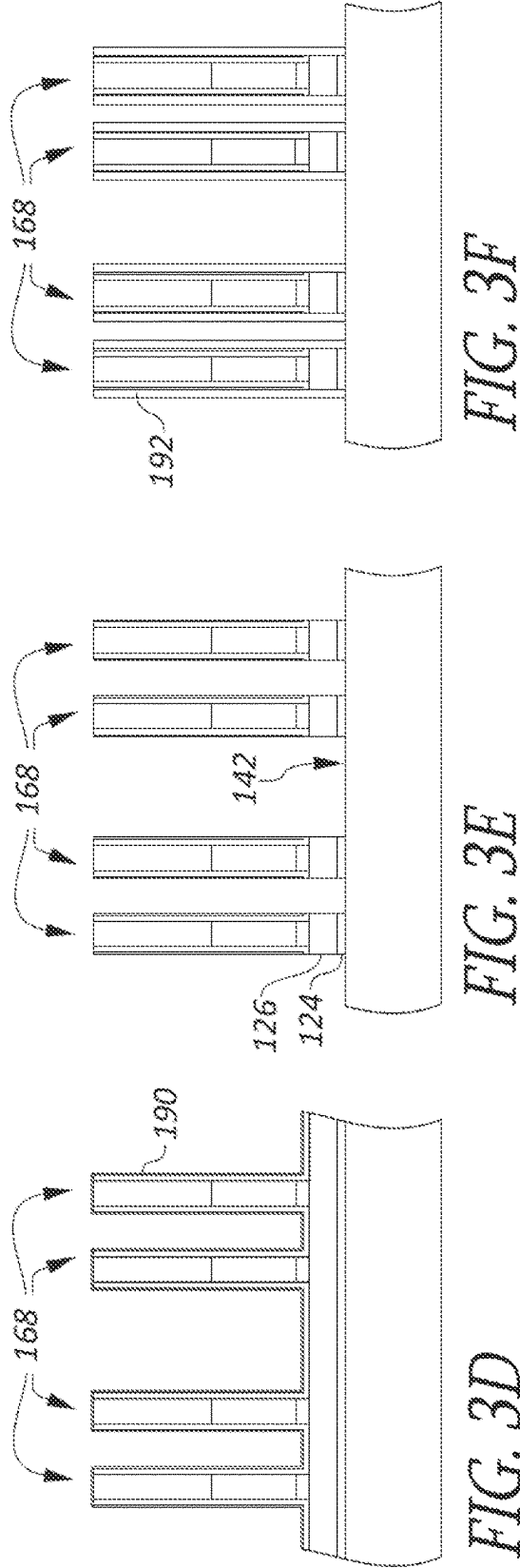
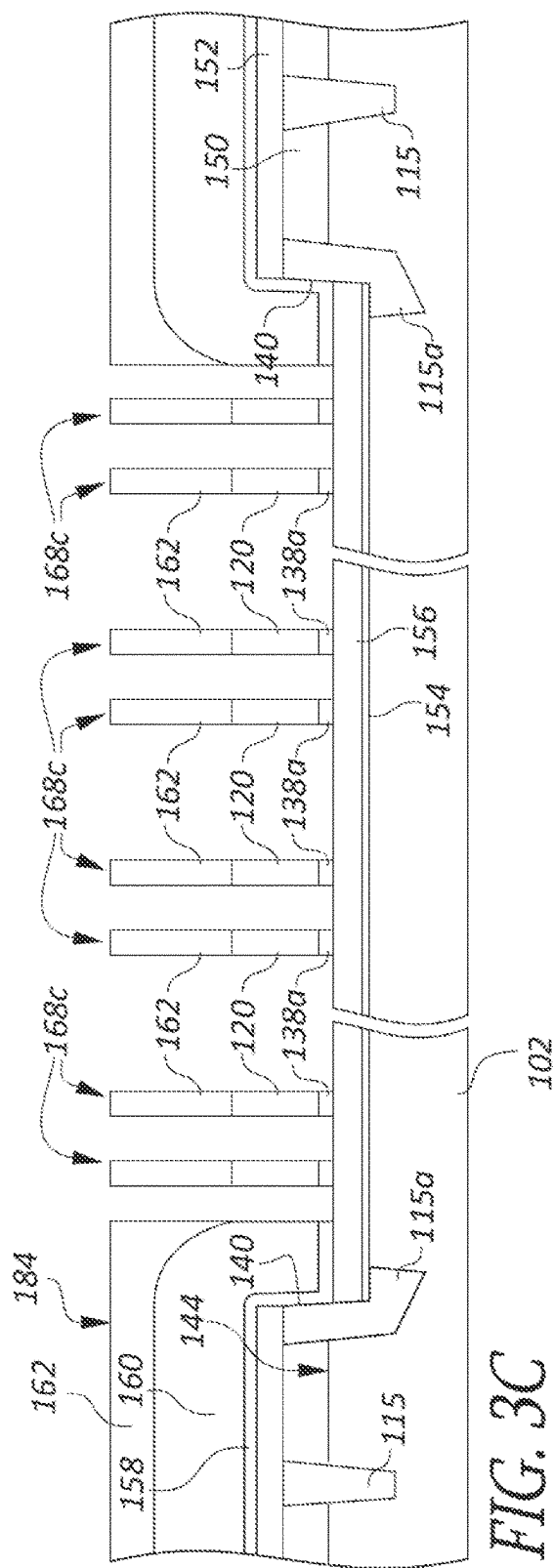


FIG. 4

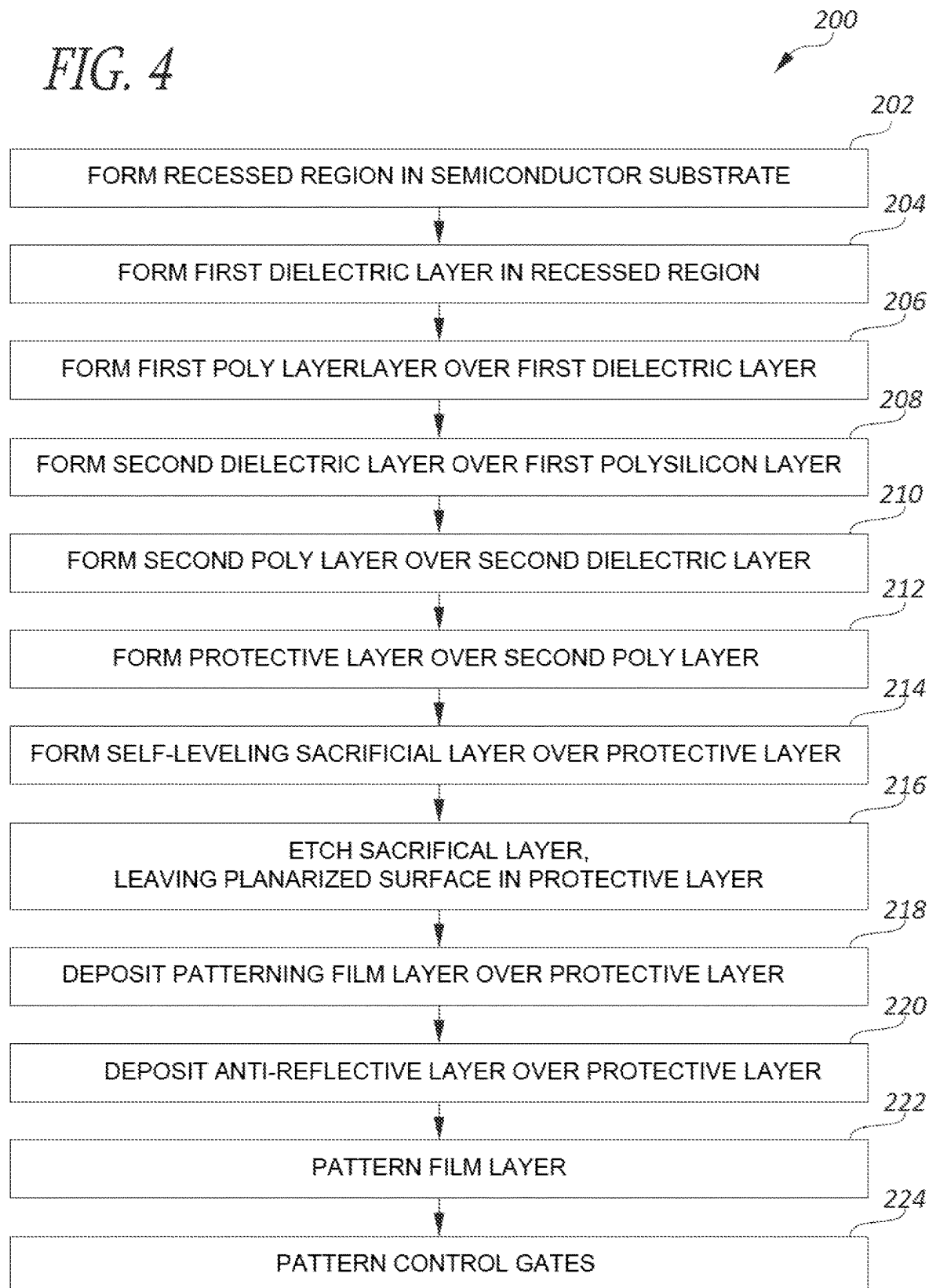
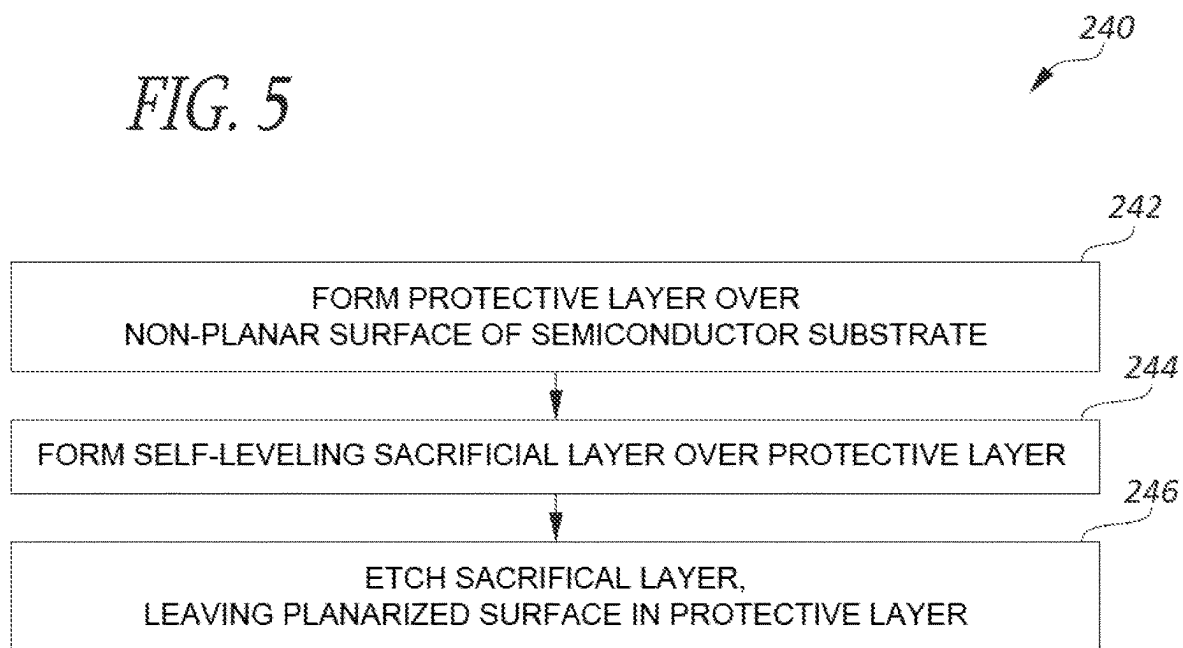


FIG. 5

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METHOD FOR IMPROVING CONTROL GATE UNIFORMITY DURING MANUFACTURE OF PROCESSORS WITH EMBEDDED FLASH MEMORY

BACKGROUND

Flash memories have some particular advantages and benefits, as compared to other types of solid-state non-volatile memory structures. Many of these advantages and benefits are related, for example, to improved read, write, and/or erase speeds, power consumption, compactness, cost, etc. Flash memories are commonly used in high-density data storage devices configured for use with cameras, cell phones, audio recorders, portable USB data storage devices—often referred to as thumb drives or flash drives—etc. Typically, in such applications, a flash memory is manufactured on a dedicated microchip, which is then coupled with another chip or chips containing the appropriate processor circuits, either together in a single package, or in separate packages configured to be electrically coupled.

Processors with embedded flash memories are a more recent development. In such devices, a flash memory array is manufactured together with logic and control circuitry on a single chip. This arrangement is often used in microcontroller units (MCU), i.e., small computer devices integrated onto single chips, which are typically designed to repeatedly perform a limited number of specific tasks. MCUs are often used in smart cards, wireless communication devices, automobile control units, etc. Integration of memory with related processing circuitry can improve processing speed while reducing package size, power consumption, and cost.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a diagrammatic side sectional view of a portion of a semiconductor device with embedded flash memory, such as, e.g., a microcontroller unit, during manufacture, according to an embodiment.

FIGS. 2A-2D are diagrammatic side sectional views of the semiconductor device of FIG. 1 at respective stages of the manufacturing process, showing, in particular, a recessed region during the formation of control and floating gates of an embedded memory array, and illustrate the source of a problem addressed by various disclosed embodiments.

FIGS. 3A-3F are diagrammatic side sectional views of the semiconductor device of FIG. 1 at respective stages of a manufacturing process, and illustrate the formation of the control and floating gates of the device 100 of FIG. 1, according to an embodiment. In particular, the process of FIGS. 3A-3F continue from the stage described above with reference to FIG. 2B, and supersede the portion of the process described with reference to FIGS. 2C-2D.

FIGS. 4 and 5 are flow chart outlining methods of manufacture, according to respective embodiments, that are consistent with processes described with reference to FIGS. 2A-2B and 3A-3F.

DETAILED DESCRIPTION

The following disclosure provides various embodiments, or examples, for implementing different features of the

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provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In the drawings, some elements are designated with a reference number followed by a letter, e.g., “704a, 704b.” In such cases, the letter designation is used where it may be useful in the corresponding description to refer to or to differentiate between specific ones of a number of otherwise similar or identical elements. Where the description omits the letter from a reference, and refers to such elements by number only, this can be understood as a general reference to any or all of the elements identified by that reference number, unless other distinguishing language is used.

Reference to a semiconductor substrate can include within its scope any elements that are formed or deposited on a substrate, unless the context clearly limits the scope further. For example, reference to planarizing a surface of a semiconductor substrate can refer to planarizing one or more layers of material deposited or otherwise formed on the actual base material of the substrate, including, for example polysilicon layers, metallic layers, dielectric layers, or a combination of materials, layers, and/or elements.

A microcontroller unit (MCU) typically includes a number of discrete devices, such as, e.g., a central processing unit (CPU) core, static random access memory (SRAM) arrays (or modules), flash memory modules, a system integration module, a timer, an analog-to-digital converter (ADC), communication and networking modules, power management modules, etc. Each of these devices, in turn, comprises a number of passive and active electronic components, such as, e.g., resistors, capacitors, transistors, and diodes. A large number of these components, particularly the active components, are based on various types of metal-oxide semiconductor field effect transistors (MOSFET), or variations thereof. In a MOSFET, conductivity in a channel region extending between source and drain terminals is controlled by an electric field in the channel region, produced by a voltage difference between a control gate and the body of the device.

FIG. 1 is a diagrammatic side sectional view of a portion of a device 100, such as, e.g., an MCU, during manufacture on a semiconductor material substrate 102, according to an embodiment. The device 100 includes a processor with an embedded flash memory array 104 and other processor

circuits 106 formed on the semiconductor substrate 102. The other processor circuits 106 include a plurality of transistors 108 configured for a variety of different functions, but which, for the purposes of the present disclosure, will be referred to as logic transistors, each of which includes a control gate 110 isolated from a channel region 112 by a gate dielectric 114. Source and drain regions, 116 are formed at opposite ends of the channel regions 112. Isolation regions 115 electrically isolate various components of the device 100 to prevent interference between different elements during operation.

The flash memory array 104 is positioned within a recessed region 140, and isolated by a shallow trench isolation (STI) region 115a. The memory array 104 includes a plurality of flash memory cells 118 that are, in many respects similar to the logic transistors 108, each having a control gate 120, a channel region 122, and a gate dielectric 124. However, each of the flash memory cells 118 also includes a floating gate 126 positioned between the control gate 120 and the gate dielectric 124. In the embodiment shown, an erase gate dielectric 128 is positioned between each floating gate 126 and a corresponding erase gate 130. Alternating source and drain regions 132, 134 are interleaved between memory cells 118 so that each memory cell shares a source 132 with an adjacent memory cell on one side and a drain 134 with an adjacent memory cell on the opposite side. Select gates 136 are positioned between the drains 134 and the control and floating gates 120, 126 of each memory cell 118. Various layers of material, 138 are shown in general outline, which are not configured to act as conductors or semiconductors in the device 100. These layers may comprise dielectrics, resist overlays, passivation layers, etch stop layers, spacers, etc., and include a dielectric layer 138a positioned between each of the floating gates 126 and the respective control gate 120 of the memory cells 118.

Because of the similarities in the structures of the logic transistors 108 and the flash memory cells 118, they would be of similar heights except for the added height of the floating gates 126 and corresponding dielectric layers 138 of the memory cells. This can be a problem because, at several points during the manufacturing process, chemical/mechanical polishing (CMP) procedures are performed to produce a substantially flattened surface for succeeding process steps. A CMP procedure that reduces the control gates 110 of the logic transistors 108 to the appropriate height can damage the taller control gates 120 of the memory cells 118.

One solution is to form the memory array 102 in a recessed region 140, in which a surface 142 of the semiconductor substrate 102 within the recessed region is reduced in height, relative to a surface 144 of the substrate outside the recessed region, by a distance about equal to the total thickness of the floating gates 126 and the dielectric 138a. This difference in height between the surfaces 142 and 144 may be termed a step height or step height differential between the surface 142 of the recessed region 140 in which memory cells of the memory array 102 are formed and the surface 144 of the periphery around the recessed region in which other components are formed. According to an embodiment, the recessed region 140 is formed via an etching procedure, in which the surface of the semiconductor substrate 102 is etched evenly across the intended recessed region to a desired depth, resulting in a substantially planar surface 142 on which the memory cells 118 are subsequently formed.

According to an alternative embodiment, a layer of semiconductor material is deposited or grown on the surface of the substrate 102 outside the intended recessed region 140,

raising the surface 144 to a desired height above the recessed surface 142. For the purposes of the present disclosure, reference to the formation of a recessed region includes any process that results in a defined region having a depth, relative to the surrounding substrate, about equal to the height difference between floating gate transistors and MOSFET transistors that do not employ floating gates.

As noted above, the logic transistors 108 operate by application of an electric field over the respective channel region 112, thereby changing the conductivity of the channel region. The electric field is produced by application of a voltage potential between the control gate 110 and the semiconductor body 102. A MOSFET can be configured either to increase or decrease conductivity when an electric field of a selected polarity is present. Typically, transistors in a logic circuit are designed to function like switches, turning on or off in response to an electric field of a selected strength, and controlling a flow of current in the channel region.

In the memory cells 118, during a write operation, electrons can be forced to tunnel through the gate dielectric 124 to the floating gate 126, where they can remain trapped indefinitely, by applying a write voltage to the control gate 120 while applying a voltage potential across the channel region 122. If there is a sufficient number of electrons trapped on the floating gate 126, the electrons can block an electric field produced by the control gate 120, preventing the control gate from acting to change conductivity in the channel region 122. Thus, the presence of electrons can be detected by applying a voltage potential across the drain and source regions 132, 134 while applying a read voltage to the control gate 120 to produce an electric field, and testing for a current flow in the channel region 122. Typically, a binary value of one is the default setting of a flash memory cell at the time of manufacture and before programming, while a binary value of zero is indicated if channel current is unaffected by a read voltage at the control gate 120. A binary zero value on a flash memory cell can be erased—i.e., returned to a one—by applying a sufficiently powerful erase voltage to the erase gate 130. This causes electrons trapped on the floating gates 126 of the memory cells 118 adjacent to the energized erase gate 130 to tunnel out through the erase gate dielectrics 128 to the erase gate 130. In practice, there are many more memory cells 118 adjacent to the erase gate 130, extending along rows lying perpendicular to the view of FIG. 1. During an erase operation, each of those memory cells is erased, hence the term flash memory.

As advances in technology enable ever smaller and more compact features, power and voltage requirements are reduced while memory density and speed increase. However, a problem that has arisen with the reduction in size is that a very small variation in gate dimensions can have a progressively increasing impact on performance, because such a variation represents a larger change, relative to the proportionately smaller nominal gate dimensions. This becomes a greater issue as technology nodes drop below 65 nm, 40 nm, and 28 nm scales.

FIGS. 2A-2D are diagrammatic side sectional views of the substrate 102 at respective stages of the manufacturing process, showing, in particular, the recessed region 140 during the formation of the control and floating gates 120, 126 of the memory array 108 of the device 100 of FIG. 1. The diagrams shown in FIGS. 2A-2D, and the corresponding description, are only isolated steps in the manufacturing process, and are not intended to provide information regarding the manufacturing process in general.

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Initially, as shown in FIG. 2A, a protective layer **150** is formed on the surface **144** of the semiconductor material substrate **102**, and can include a plurality of layers of, for example, silicon oxide, silicon nitride, etc. Isolation regions **115** are formed in the substrate **102**, and a hard mask layer **152** is deposited. The recessed region **140** is then formed in an anisotropic etch process. An oxide layer **154** is formed on the substrate surface **142** within the recessed region **140** and a polysilicon layer **156** is grown over the oxide layer. The oxide and polysilicon layers **154**, **156** comprise the materials from which the gate dielectrics **124** and floating gates **126** of the memory array **104** will eventually be formed.

Proceeding to the stage shown in FIG. 2B, a dielectric layer **158** is deposited over the substrate **102**, followed by a polysilicon layer **160** and a protective layer **162**. The dielectric and polysilicon layers **158**, **160** comprise the materials from which the dielectric layers **138a** and control gates **120** of the memory array **104** will eventually be formed, while the protective layer **162** will be patterned with the dielectric and polysilicon layers to form protective caps over the gate stacks to protect the control and floating gates during later process steps. According to an embodiment, the protective layer **162** is a hard mask material—typically an oxide, nitride, amorphous carbon, a combination of such materials, etc. The term hard mask material refers to a layer (or layers) of material that is substantially resistant to selected ones of the processes that will be employed following the formation of the mask. The resistance to selected ones of the processes that will be employed following the formation of the mask can be a result of the composition of the hard mask material or a combination of the composition of the hard mask material and the thickness of the hard mask material. The thickness of the hard mask material can vary and embodiments of the present disclosure are not limited to the specific thicknesses described below. In various embodiments, the hard mask material includes a silicon nitride (SiN) layer of 400 angstroms to 500 angstroms (Å), a silicon dioxide (SiO₂) layer of 1000 Å to 1200 Å, and a SiN layer of 700 Å to 900 Å. In some embodiments the hard mask material includes a SiN layer of 440 Å, a SiO₂ layer of 1100 Å, and a SiN layer of 800 Å. In these embodiments, the total thickness of the hard mask material formed by these layers is 2340 Å which is greater than a depth of the recessed region **140**, where the recessed region has a depth of greater than 300 Å. The thickness of these layers is based on an etching selectivity of a subsequent etching process described below with reference to FIG. 2D. In various embodiments, the total thickness of the hard mask material formed by these layers ranges from 2100 Å to 2600 Å and is chosen to be greater than a depth of the recessed region **140**, where the recessed region has a depth of greater than 300 Å.

In FIG. 2C, a bottom anti-reflection coating (BARC) **164** is applied over the protective layer **162**, followed by a patterning film layer **166**. The patterning film layer **166** will be patterned to produce an etch mask, with the BARC **164** positioned between the substrate and the patterning film layer to prevent damage to the film layer caused by reflections from the surface of the substrate **102** during the patterning of the film. The BARC is typically applied via a spin-coating process, while, depending upon the type of patterning film, the patterning film **166** may be applied via any of several processes, including, for example, deposition processes such as thermal deposition or chemical vapor deposition (CVD). As shown in FIG. 2C, the BARC **164** tends to deposit more thickly around the edges (**164b**) of the recessed region **140** than in the center (**164a**). If the pat-

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terned film is applied by spin-coating, it too will likely be thicker at the edges (**166b**) than in the middle of the recessed region (**166a**).

Turning finally to FIG. 2D, after patterning of the patterning film layer **166**, an etching process is performed, producing a plurality of gate stacks **168**, each having a protective cap **170**, a control gate **120**, and a dielectric layer **138a**. Remnants of the oxide layer **158**, the polysilicon layer **160**, and the protective layer **162** remain around the periphery of the recessed region **140**. The gate dielectrics **124** and floating gates **126** will be etched from their parent layers **154**, **156** during a later step, after additional process steps are performed, including the deposition of further dielectric layers.

As shown in FIG. 2D, the gate stacks **168a** near the center of the recessed region **140** are narrower than the gate stacks **168b** closer to the edges of the region. This is caused by the thicker BARC and patterning film layer portions **164b** near the edges of the recessed region **140**, as described above with reference to FIG. 2C. Typically, BARC material is considered to be conformal, and the impact of small variations in thickness caused by surface features of the substrate has been negligible. As scales continue to drop, the impact of small variations in thickness caused by surface features of the substrate may become a significant problem, particularly at technology nodes below, e.g., the 65 nm, 40 nm, and 28 nm scales. This is because the dimensions of the control and floating gates **120**, **126** have a direct impact on critical operating characteristics of the devices, such as read and write speeds, program and erase state voltage and current, and power consumption, etc.

In order to utilize the memory cells **118** near the edges of the recessed region **140**, the operating parameters of the entire memory array **108** can be modified, to ensure that data is not lost or corrupted because the memory cells near the edges of the recessed region are not properly written or erased. Modifying the operating parameters of the entire memory cell in this manner can result in less than optimal performance of the entire array. One alternative is to leave the cells nearest the edges of the recessed region **140** inactive—or leave the edge region empty—but this will result in a loss of capacity. The reduced total capacity may be minimal if all of the memory cells of a device are in a single array, but many MCU devices are designed to place smaller memory arrays adjacent to circuits that will be using them, to increase throughput speed. In such devices, smaller memory arrays are positioned at multiple locations on the device. Consequently, the total area of the edge regions is much larger than in a single array, so the lost capacity is also much greater.

The inventors have also recognized that the problem could be eliminated if the BARC were deposited on a substantially planar surface. However, planarizing the surface at the process stage shown in FIG. 2B is problematic. A CMP process would have a tendency to produce dishing over the recessed region **140**, and other planarizing processes are inapplicable or would require many additional process steps.

Embodiments described hereafter with reference to FIGS. 3A-3F reduce or eliminate the BARC layer thickness variations and the associated variation in gate uniformity described above. FIGS. 3A-3F are diagrammatic side sectional views of the substrate **106** at respective stages of a manufacturing process, according to an embodiment, showing formation of the control and floating gates **120**, **126** of the memory array **108** of the device **100** of FIG. 1. In particular, the process of FIGS. 3A-3F continue from the

stage described above with reference to FIG. 2B, and replace the portion of the process described with reference to FIGS. 2C-2D. Thus, proceeding from the stage of FIG. 2B, FIG. 3A shows the deposition of a sacrificial layer **180** over the protective layer **162**. According to embodiments of the present disclosure, the sacrificial layer **180** is a photoresist layer that is applied in a spin coating process, with a thickness that is sufficient to completely fill the depression in the protective layer **162** over the recessed region **140**. The sacrificial layer **180** is self-leveling, meaning that, as applied, its upper surface **182** is substantially planar, without requiring further processing. A thickness of the sacrificial layer **180** is about 1000 Å per spin-coating process in one embodiment of the present disclosure. The thickness of the sacrificial layer **180** is not limited to being about 1000 Å per spin-coating process and can be greater or less than 1000 Å per spin-coating process. The sacrificial layer **180** is not limited to being a photoresist layer, and may be other flowable materials in further embodiments of the present disclosure. In addition, the flowable material of the sacrificial layer **180** is not limited to being applied through a spin coating process, with other suitable processes being utilized to apply the flowable material in further embodiments.

As shown in FIG. 3B, a non-selective etch process is performed, etching the surface of the substrate back into the protective layer **162**, while completely removing the sacrificial layer **180**. The chemistry of the etching process is selected such that the etch rates of the sacrificial layer **180** and of the protective layer **162** are substantially equal. Consequently, the etching process proceeds evenly across the surface of the substrate, leaving a planarized face **184** on the surface of the protective layer **162**. A patterning film layer **186** and a BARC layer **188** are then deposited at uniform thicknesses over the planarized surface **184** of the protective layer **162**. The BARC layer **188** is typically a suitable flowable material and is formed through a spin-coating process while the patterning film layer **186** is formed through a suitable process such as a chemical vapor deposition (CVD) process, as discussed above for corresponding layers in relation to FIGS. 2A-2D. The upper surface **182** is said to be substantially planar in embodiments of the present disclosure, where the term "substantially" indicates the upper surface is sufficiently planar to enable the subsequent formation of the patterning film layer **186** and BARC layer **188** having reduced thickness variations to thereby enable the formation of control gates having uniform thicknesses as described above.

Other processes may be utilized to remove the sacrificial layer **180** and to partially remove the protective layer **162**. The portion of the protective layer **162** remaining after the etching of the surface of the substrate to remove the sacrificial layer **180** and partially remove the protective layer depends on the specific etching process utilized. In one embodiment, the etching process partially removes the protective layer **162** to leave a remaining portion of the protective layer having a depth of 1500-2000 angstroms. The described embodiments of the present disclosure, which apply the sacrificial layer **180** through spin coating and remove the sacrificial layer and partially remove the protective layer, improve the uniformity of the BARC layer **188** deposited across the center and periphery portions of the flash memory array **104** (FIG. 1). This uniformity of the BARC layer **188** enables the formation of gate stacks **168c** (FIG. 3C) having uniform thicknesses for the gate stacks in the center and in the periphery portions of the flash memory array **104**.

According to an embodiment, the etch of the sacrificial layer **180** proceeds directly following the deposition and—if necessary—planarizing of the sacrificial layer, without any intervening process steps. According to another embodiment, one or more process steps are performed between the deposition of the sacrificial layer **180** and its subsequent removal in the etch procedure. These intervening process steps can include processes that are unrelated to the formation of the memory array **104**.

Proceeding to FIG. 3C, the patterning film layer **188** is patterned to form an etch mask, and the protective, polysilicon and dielectric layers **162**, **160**, **158** are etched to form gate stacks **168c**. In contrast with the gate stacks **168a** and **168b** of FIG. 2D, the gate stacks **168c** of FIG. 3C are substantially equal in width, owing to the uniform thickness of the BARC layer **188** and the patterning film layer **186**, which, in turn, is due to the planarized face of **184** of the protective layer **162**.

FIGS. 3D-3E are diagrammatic side sectional views of the substrate **106**, showing a small portion of the recessed region **140**, and illustrating the manufacturing process through a stage in which the gate stacks **168** of the memory array are substantially complete. In FIG. 3D, one or more dielectric layers **190** are deposited over the gate stacks **168**. In FIG. 3E, the floating gates **126** and gate dielectrics **124** are formed in an etching process in which the gate stacks **168** act as self-aligned masks. Finally, in FIG. 3F, an oxide is formed over the gate stacks **168** and etched to leave protective dielectric layers **192** on the sides of the gate stacks.

FIGS. 4 and 5 are flow chart outlining methods of manufacture, according to respective embodiments, that are consistent with processes described above with reference to FIGS. 2A-2B and 3A-3F.

FIG. 4 outlines a method **200**, according to an embodiment, in which, at step **202**, the recessed region **140** is formed in the semiconductor substrate **102**. The first dielectric layer **154** is formed over the substrate **102** within the recessed region **140** in step **204**, and the first polysilicon layer **156** is formed over the first dielectric layer **154** in step **206**. In step **208**, the second dielectric layer **158** is formed, followed by formation of the second polysilicon layer **160** in step **210**. In step **212**, the protective layer **162** is formed over the second polysilicon layer **160**. In step **214**, the sacrificial layer **180** is formed over the protective layer **162**, self-leveling to form the planar upper surface **182**. In step **216**, in a non-selective etch process the sacrificial layer **180** is removed, together with a portion of the protective layer **162**, planarizing the exposed surface **184** of the protective layer. The patterning film layer **186** is deposited on the surface **184** of the protective layer **162** and the anti-reflective coating **188** is deposited over the substrate, in respective steps **218** and **220**, and the patterned film layer **186** is patterned in step **222**. Finally, a plurality of control gates **120** are defined in an etch process regulated by the patterned film layer, in step **224**.

FIG. 5 is a flow chart outlining a method **240** for providing a planarized surface in a manufacturing process, according to another embodiment. In step **242**, the protective layer **162** is deposited over a non-planar surface on the semiconductor substrate **102**. In step **244**, the self-leveling sacrificial layer **180** is formed over the protective layer **162**, forming a planar upper surface **182**. In step **246**, a non-selective etch procedure is performed in which the entirety of the sacrificial layer **180** and a portion of the protective layer **162** are removed to leave a planarized surface **184** of the remaining portion of the protective layer **162**.

The embodiments shown and described herein provide improvements to a process for manufacturing microelectronic devices that include embedded flash memory arrays. According to various embodiments, prior to defining control and floating gate stacks of the memory arrays in recessed regions on a semiconductor substrate, a planar surface is provided on the substrate over layers of material from which the gate stacks are to be formed, for deposition of an anti-reflective coating and of a patterning film at an even thickness. This is beneficial because the anti-reflective coating, in particular, has a tendency to vary in thickness when it is deposited over a non-planar surface, which in turn, can contribute to non-uniformity of control and floating gate dimensions. The dimensions of the control and floating gates have a direct impact on critical operating characteristics of a memory device, such as read, write, and erase speeds, program and erase state voltage and current levels, power consumption, etc. If gate dimensions vary within a memory array, a typical practice is to operate the entire array based on the operating characteristics of the weakest cells. Thus, significant variation in dimensions is a problem because, if even a small percentage of the cells in an array require higher voltages and/or longer read and write times, the entire array is operated at the same levels, resulting in a loss of efficiency and speed for the entire array. By providing the planar surface, non-uniformity in gate dimensions of the memory arrays is reduced or eliminated. This, in turn, results in arrays with higher overall speed and efficiency.

According to an embodiment, the improvements include forming a protective layer over a non-planar surface of a semiconductor substrate, in particular, for example, over a recessed region of the semiconductor substrate, in which an embedded memory array is to be formed. A sacrificial layer is then deposited over the protective layer, to a depth sufficient to permit formation of a substantially planar surface on the sacrificial layer. The sacrificial layer is then etched to a depth that removes the sacrificial layer and leaves a planar surface formed in the protective layer.

According to another embodiment, the method includes forming a recessed region in a semiconductor substrate, and forming gate materials for a plurality of floating gates and control gates of a memory array, in layers over the recessed region. A protective layer is then formed over the gate materials in the recessed region, and planarized to improve uniformity of gate dimensions. An etch mask is formed over the planarized protective layer, and gate stacks of the memory array are formed in the recessed region, by etching the gate materials. According to an embodiment, planarizing the protective layer includes depositing a self-leveling sacrificial layer over the protective layer to produce a substantially planar surface, then etching the surface at an even rate, and to a depth sufficient to remove the sacrificial layer, which results in a planar surface on the protective layer.

According to another embodiment, the method includes forming a protective layer over a stack of gate-material layers in a recessed region of a semiconductor substrate. A self-leveling sacrificial layer is then deposited over the protective layer to a depth sufficient to produce a planarized surface of the semiconductor substrate, and the sacrificial layer is etched back at an even rate to a depth sufficient to produce a substantially planar surface on the protective layer.

Ordinal numbers, e.g., first, second, third, etc., are used in the claims according to conventional claim practice, i.e., for the purpose of clearly distinguishing between claimed elements or features thereof, etc. Ordinal numbers may be assigned arbitrarily, or assigned simply in the order in which

elements are introduced. The use of such numbers does not suggest any other relationship, such as order of operation, relative position of such elements, etc. Furthermore, an ordinal number used to refer to an element in a claim should not be assumed to correlate to a number used in the specification to refer to an element of a disclosed embodiment on which that claim reads, nor to numbers used in unrelated claims to designate similar elements or features.

While the method and process steps recited in the claims may be presented in an order that corresponds to an order of steps disclosed and described in the specification, except where explicitly indicated, the order in which steps are presented in the specification or claims is not limiting with respect to the order in which the steps may be executed.

The abstract of the present disclosure is provided as a brief outline of some of the principles of the invention according to one embodiment, and is not intended as a complete or definitive description of any embodiment thereof, nor should it be relied upon to define terms used in the specification or claims. The abstract does not limit the scope of the claims.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A method, comprising:

planarizing a protective layer over gate materials overlying a first region in a substrate, the first region being adjacent a second region, the second region having thickness exceeding that of the first region, the planarizing including:

forming a first planarized surface by planarizing a sacrificial layer over the protective layer; and

forming a second planarized surface of the protective layer by etching the first planarized surface of the sacrificial layer at an even rate across the first region, the second planarized surface being above a polysilicon layer of the gate materials;

forming an etch mask layer over the second planarized surface; and

forming control gate stacks in the first region by etching the gate materials and the protective layer, the protective layer forming protective caps over the control gate stacks.

2. The method of claim 1, wherein the etching the first planarized surface partially removes the protective layer to leave a remaining portion of the protective layer having a depth of 1500-2000 angstroms.

3. The method of claim 1, further comprising forming a first dielectric layer on opposing sidewalls of the control gate stacks and over a floating gate layer.

4. The method of claim 3, further comprising forming floating gate stacks by etching the floating gate layer.

5. The method of claim 4, further comprising forming dielectric layers by etching a second dielectric layer underlying the floating gate layer.

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6. The method of claim 1, wherein the forming the protective layer comprises forming a protective layer that includes a first nitride layer, an oxide layer and a second nitride layer.

7. The method of claim 6, wherein the forming the protective layer includes forming a protective layer having thickness greater than depth of the recessed region.

8. The method of claim 7, wherein the forming the protective layer includes forming the protective layer having thickness in a range of 2100 angstroms to 2600 angstroms.

9. The method of claim 8, wherein the forming the protective layer includes forming a first silicon nitride layer of 400 angstroms to 500 angstroms, a silicon dioxide layer of 1000 angstroms to 1200 angstroms, and a second silicon nitride layer of 700 angstroms to 900 angstroms.

10. A method, comprising:

forming a first region adjacent a second region, the first region being recessed relative to the second region;

forming a first dielectric layer in the first region;

forming a floating gate layer in the first region on the first dielectric layer;

forming a second dielectric layer in the first region and the second region on the floating gate layer;

forming a control gate layer over the floating gate layer and extending to the second region;

forming a hard mask layer over the control gate layer;

forming a sacrificial layer over the hard mask layer, the sacrificial layer having a first planar surface;

forming a planarized surface of the hard mask layer by removing the sacrificial layer and the hard mask layer at substantially equal rates in an etch process;

forming an etch mask layer over the hard mask layer to a substantially uniform thickness across the first region; and

forming gate stacks in the first region having substantially uniform width by etching floating gate layer and the control gate layer under the etch mask layer, wherein the forming the gate stacks includes:

forming protective caps and control gates by etching through the hard mask layer, the control gate layer and the second dielectric layer;

forming a third dielectric layer on sides of the protective caps and control gates; and

forming floating gates by etching through the floating gate layer and the first dielectric layer while the third dielectric layer is present.

11. The method of claim 10, wherein the forming the gate stacks includes forming floating gates from the floating gate layer, control gates from the control gate layer and protective caps from the hard mask layer.

12. The method of claim 10, further comprising forming a fourth dielectric layer on sides of the third dielectric layer, the floating gates and the first dielectric layer.

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13. The method of claim 10, wherein:

forming the first dielectric layer includes forming an oxide layer on a surface of the substrate in the first region; and

forming the floating gate layer includes growing a polysilicon layer on the oxide layer.

14. The method of claim 10, wherein forming the etch mask layer includes forming a bottom anti-reflective coating (BARC) layer.

15. A method, comprising:

forming a protective layer having a depression over a first region of a substrate, the protective layer being over gate materials of a memory array in the first region and over a periphery around the first region, the first region being recessed relative to the periphery;

forming a self-leveling layer to a level above an upper plane of the depression, wherein an upper surface of the self-leveling layer is substantially planar as the self-leveling layer is applied;

forming a planarized surface on the protective layer by etching the self-leveling layer at an even rate across the first region to a depth sufficient to remove the self-leveling layer and a portion of the protective layer; and

forming gate stacks in the first region by etching a floating gate layer and a control gate layer under an etch mask layer, wherein the forming the gate stacks includes:

forming protective caps and control gates by etching through the protective layer and the control gate layer;

forming a third dielectric layer on sides of the protective caps and control gates; and

forming floating gates by etching through the floating gate layer while the third dielectric layer is present.

16. The method of claim 15, wherein the forming a self-leveling layer includes forming a flowable material layer by a spin-coating process.

17. The method of claim 16, wherein the forming a flowable material layer includes forming a photoresist layer.

18. The method of claim 17, wherein the forming a photoresist layer includes forming a photoresist layer to thickness of at least 1000 angstroms per spin-coating process.

19. The method of claim 15, wherein the etching a floating gate layer and a control gate layer under an etch mask layer includes etching the floating gate layer and the control gate layer under a bottom anti-reflective coating (BARC) layer.

20. The method of claim 15, further comprising forming a dielectric layer on sidewalls of the floating gates and the third dielectric layer.

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